

Bihar Engineering University, Patna

B.Tech 3rd Semester Examination, 2024

Course: B.Tech
Code: 100313

Subject: Object Oriented Programming using C++

Time: 03 Hours
Full Marks: 70

Instructions:-

- (i) The marks are indicated in the right-hand margin.
- (ii) There are NINE questions in this paper.
- (iii) Attempt FIVE questions in all.
- (iv) Question No. 1 is compulsory.

Q.1 Choose the correct option / answer the following (Any seven question only):

[2 x 7 = 14]

- (a) What is the default access specifier for class members in C++?
- (i) public
 - (ii) protected
 - (iii) ☒ private
 - (iv) internal
- (b) What will be the output of the following code?
- ```
#include <iostream>
class Test {
public:
 Test() { std::cout << "Constructor"; }
 ~Test() { std::cout << "Destructor"; }
};
int main() {
 Test t;
 return 0;
}
```
- (i) Constructor
  - (ii) ☒ ConstructorDestructor
  - (iii) Destructor
  - (iv) DestructorConstructor
- (c) What will be the output of the following code?
- ```
#include <iostream>
void display(int x) { std::cout << "Int"; }
void display(double x) { std::cout << "Double"; }
int main() {
    display(5.5f);
}
```
- (i) Int
 - (ii) Float
 - (iii) ☒ Double
 - (iv) ☒ Error
- (d) What is the output of the following code snippet?
- ```
int arr[] = {1, 2, 3};
int* p = arr;
cout << *(p + 1);
```
- (i) 1
  - (ii) 3
  - (iii) ☒ 2
  - (iv) Error
- (e) What is the output of the following code snippet?
- ```
string s = "C++";
s += " Rocks";
cout << s;
```
- (i) C++
 - (ii) ☒ C++ Rocks
 - (iii) Rocks
 - (iv) Error

- (f) Which of the following statement is incorrect?
- | | |
|---|--|
| (i) A static member function cannot be declared as virtual function | (iii) A static member function cannot access the "this pointer" |
| (ii) Static and non-static functions with identical signatures cannot coexist in the same class in C++. | (iv) A non static member function can access only non static data members of a class |

- (g) Identify the correct function template syntax.

- | | |
|---|--|
| (i) <code>template function<T> void show(T t);</code> | (iii) <code>template<class> show(T t);</code> |
| (ii) <code>function template<T> show(T t);</code> | (iv) <code>template<typename T> void show(T t);</code> |

- (h) What will be the output of this code?

```
#include <iostream>
class Demo {
public:
    void show() { std::cout << "Hello"; }
};
int main() {
    Demo d, *ptr = &d;
    ptr->show();
}
```

- | | |
|------------|------------|
| (i) Hello | (iii) show |
| (ii) Error | (iv) None |

- (i) What happens when a derived class object is assigned to a base class pointer?

```
class Base { };
class Derived : public Base { };
int main() {
    Derived d;
    Base* b = &d;
}
```

- | | |
|------------------|-------------------------|
| (i) Error | (iii) Requires template |
| (ii) Not allowed | (iv) Upcasting; allowed |

- (j) What is the output of this code?

```
#include <iostream>
class A {
public:
    ~A() { std::cout << "Destructor "; }
};
int main() {
    try {
        A a;
        throw 1;
    } catch (...) {
        std::cout << "Caught";
    }
}
```

- | | |
|------------------------|--------------|
| (i) Caught Destructor | (iii) Caught |
| (ii) Destructor Caught | (iv) Error |

- Q7 (a) Explain the features of Object-Oriented Programming (OOP). How does OOP differ from the procedural programming paradigm? Also, list the advantages of using OOP in C++. [7]
- (b) Differentiate between while, do-while, and for loops in C++. Write a program to print the Fibonacci series using any loop. Hint: Fibonacci series: 0, 1, 1, 2, 3, 5, 8, 13, 21.... [7]

- Q.3** (a) Write a C++ program using function overloading to calculate the area of a square, rectangle, and circle using three overloaded area() functions. Demonstrate all three in main() and explain how function overloading works. [7]
- (b) Write a C++ program using the string class to input two strings, concatenate them, compare them, and find the length of the resulting string. [7]
- Q.4** (a) Differentiate between a struct and a class in C++. Explain with examples how data hiding and default access specifiers differ between the two. [7]
- (b) Write a C++ program to define a class Account with data members like accountNumber, holderName, and balance. Include a default constructor to initialize default values, a parameterized constructor to assign specific values, and a destructor to display a message when the object is destroyed. In the main() function, create objects using both constructors. Also, briefly explain the role of constructors and destructors in C++. [7]
- Q.5** (a) Write a program to overload the + operator to add two complex numbers using a class Complex. Also, discuss the rules and limitations of operator overloading. [7]
- (b) What is a function template in C++? Write a template function maximum() that returns the larger of two values. Demonstrate its use with integers, floats, and characters. [7]
- Q.6** (a) What is a virtual base class in C++? Explain its role in resolving the diamond problem with an example involving classes A, B, C, and D, where D inherits from both B and C, and B and C inherit from A. [7]
- (b) Explain single and multiple inheritance in C++ with suitable examples. Write a program to demonstrate both types of inheritance. Also, discuss how ambiguity is handled in multiple inheritance. [7]
- Q.7** (a) Write a C++ program to demonstrate the use of an abstract class with a pure virtual function. Explain how this supports runtime polymorphism. [7]
- (b) What is a void pointer in C++? Explain its use with an example. Also, demonstrate how a void pointer can be typecast to access different data types. [7]
- Q.8** (a) Write a C++ program to demonstrate a pointer to a class and a pointer to an object. Create a class Student and use a pointer to access its member functions. Explain the concept of dynamic object access using pointers. [7]
- (b) Explain how arrays and pointers are related in C++. Write a program to process an array using pointers (e.g., calculate the sum of elements). Discuss pointer arithmetic with arrays. [7]
- Q.9** (a) What is exception handling in C++? Explain the purpose and working of the try, throw, and catch keywords with proper syntax. Also, briefly explain the concept of stack unwinding in the context of exceptions. [7]
- (b) Write a program to handle division by zero using exception handling. Also, describe how control is transferred during an exception. [7]